

Abstract

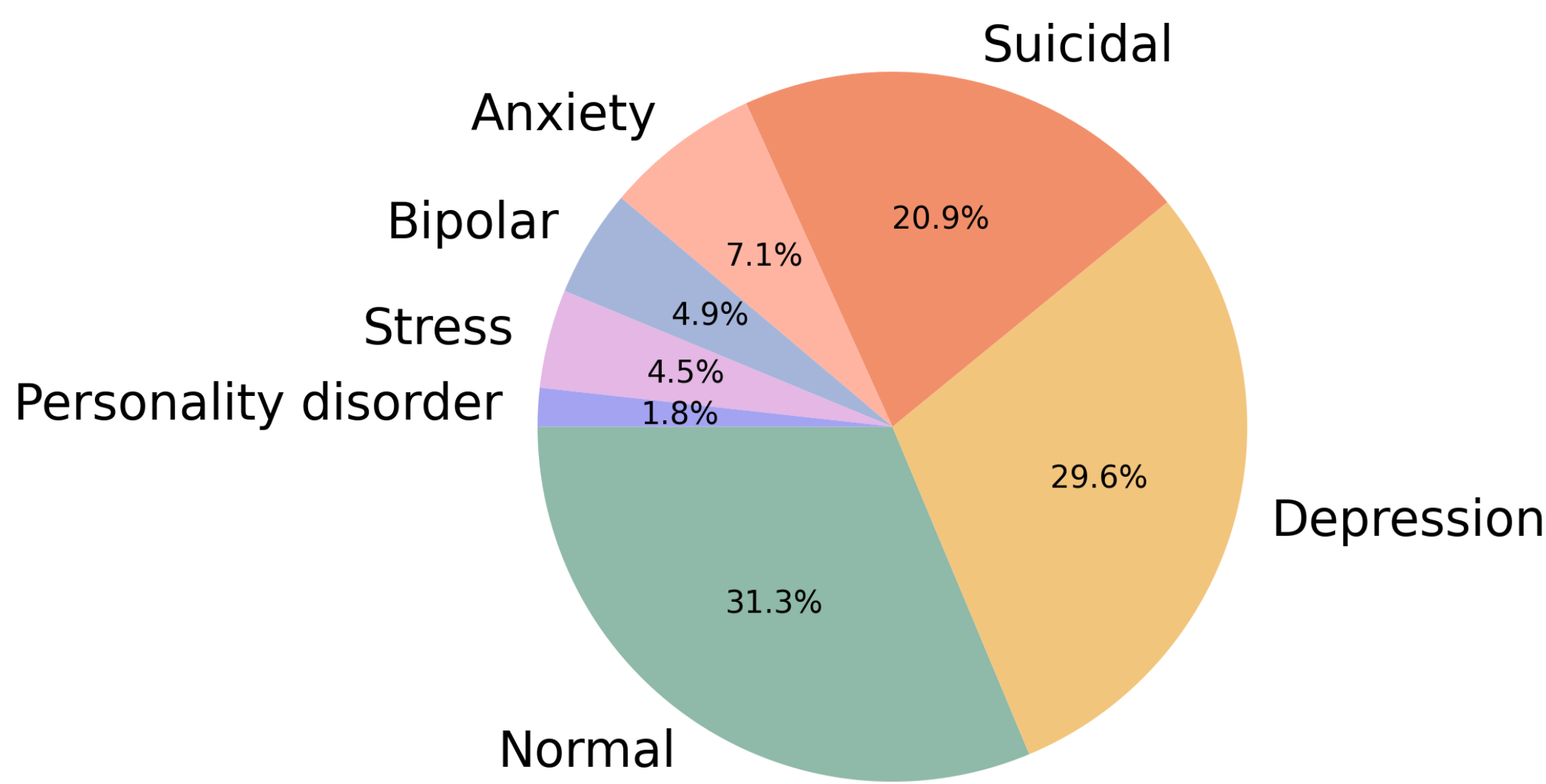
- Developed a comparative NLP framework to classify mental-health related text and systematically analyze how different language models capture emotional and psychological cues expressed in user-generated content.
- Results show a clear performance hierarchy, with the transformer based approach producing the strongest performance and most reliably identifying nuanced mental-health cues.

Goal of the Project

- To build an NLP system that identifies psychological signals in short user-generated text and evaluate how traditional, neural, and transformer-based approaches perform on mental-health text classification.
- To determine which modeling strategy is best suited for reliable, real-world mental-health detection.

Dataset

- Used a large mental-health text corpus containing **50,988 cleaned posts** from real users.
- Includes seven clinically relevant categories, with clear **class imbalance** across conditions.
- Exploratory analysis revealed:
 - Distinct linguistic patterns across categories (e.g., “hopeless”, “empty” for Depression vs. “worried”, “panic” for Anxiety).
 - Emotional cues captured through **emoji/emoticon tags**, useful for detecting subtle expressions.
 - Rare categories (e.g., Personality disorder) show limited vocabulary diversity, increasing modeling difficulty.



Methodology

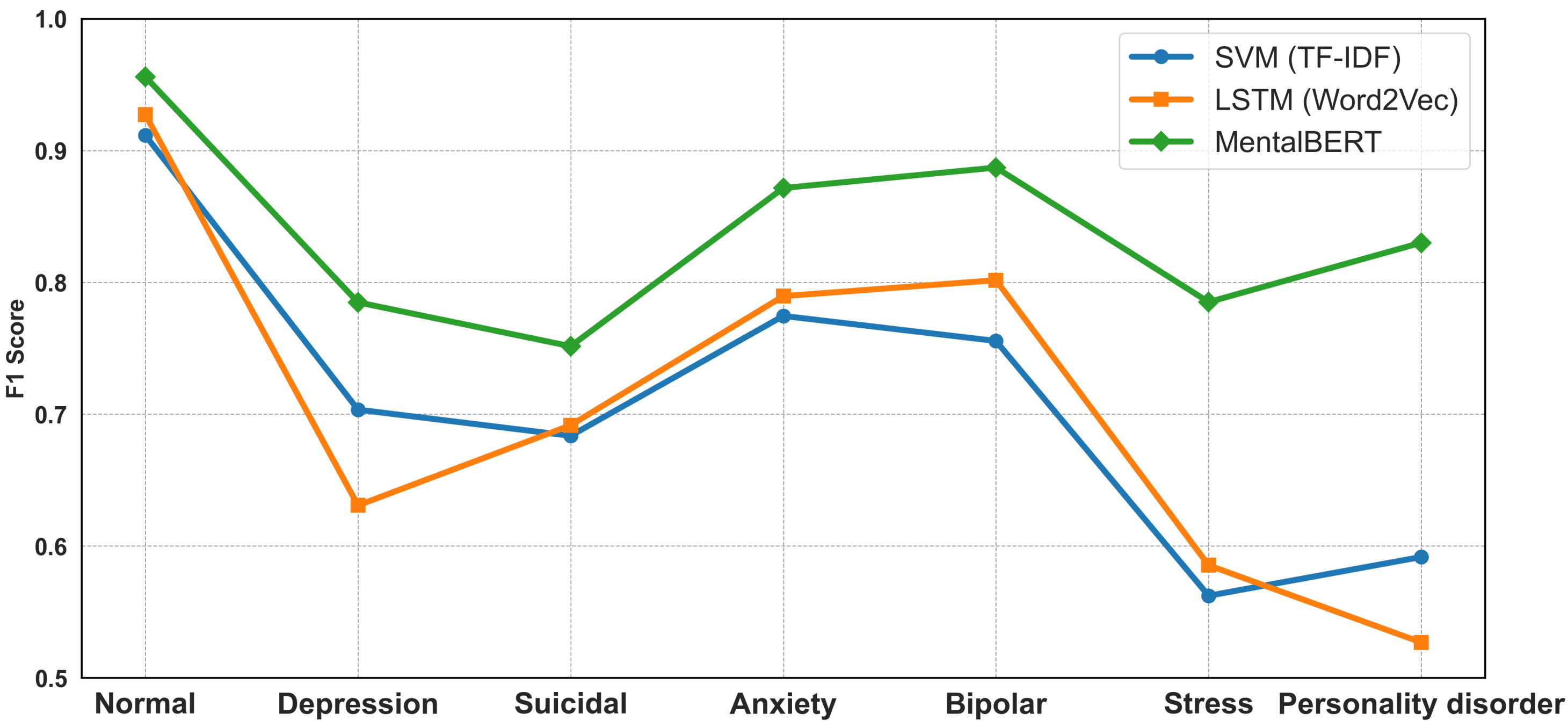
- Traditional baselines were tested: **Multinomial Naive Bayes** performed poorly on imbalanced, nuanced classes, so **TF-IDF + SVM** was chosen as a stronger and more reliable baseline.
- Built a **Word2Vec + LSTM** model to capture sequential patterns and semantic relationships. Both TF-IDF and Word2Vec require a **fully preprocessed dataset** to obtain consistent and meaningful features.
- Fine-tuned **MentalBERT**, a domain-adapted transformer pretrained on mental-health text. Transformers operate on **raw, minimally processed text** so their tokenizer can preserve subword patterns and contextual cues without aggressive cleaning.
- All models were trained and evaluated on the same dataset split using *Accuracy, Macro F1, per-class reports, and confusion matrices* to ensure a fair and consistent comparison.

Design Aspect	TF-IDF + SVM	Word2Vec + LSTM	MentalBERT
Input Required	Fully cleaned text	Tokenized + cleaned text	Raw text
Bag-of-words features only	✓		
Learns word embeddings		✓	✓
Uses sequence / word order		✓	✓
Deep contextual understanding			✓
Domain-adapted to mental health			✓
Computational cost	Low	Medium	High

Results

- Performance was evaluated on the held-out test set using **Test Accuracy** and **Macro F1** across all three approaches.
- 5-fold cross-validation for the TF-IDF + SVM baseline showed **stable performance** (Accuracy $\approx 0.775 \pm 0.001$, Macro F1 $\approx 0.728 \pm 0.005$), confirming it as a strong and reliable traditional baseline.

Model	Test Accuracy	Macro F1
TF-IDF + SVM	0.7664	0.7118
Word2Vec + LSTM	0.7345	0.6857
MentalBERT	0.8423	0.8380



- The per-class F1 comparison shows that **MentalBERT** consistently outperforms both TF-IDF + SVM and Word2Vec + LSTM, particularly on challenging categories such as *Stress* and *Personality disorder*.

Analysis

- The results show a clear model hierarchy, with the **transformer-based approach performing best overall**, while TF-IDF + SVM provides a strong baseline and Word2Vec + LSTM helps in classes where sequential patterns matter.
- Analysis of confusion patterns shows consistent overlap between *Depression–Suicidal* and *Anxiety–Stress*, indicating that semantically similar categories remain challenging across all models.

Class	Difficulty	Evidence
Normal	Easy	All models achieve F1 > 0.90; clear, non-clinical language.
Depression	Medium	Overlaps with Suicidal; SVM/LSTM weaker, MentalBERT clearly improves recall.
Suicidal	Med-hard	Needs contextual cues rather than explicit keywords; MentalBERT performs best.
Anxiety	Easier for BERT	Emotional language (“panic”, “worried”) captured well by transformer.
Bipolar	Medium	Mixed mood/state descriptions; LSTM competitive, MentalBERT strongest overall.
Stress	Hardest	Strong overlap with Anxiety; all models show lower recall.
Personality disorder	Very hard	Very few samples; only MentalBERT learns stable patterns.

Limitations and Future Work

- Dataset imbalance limits performance on minority classes ; more data or augmentation could improve stability.
- Future work can explore larger or specialized transformer models and richer linguistic signals (syntax, discourse, sentiment flow) to enhance contextual understanding.
- Real-world deployment requires explainability, bias evaluation, and safety checks—especially for high-risk predictions like *Suicidal* cases.

References

[1] Sayan, B.T.W.I.T. (2024). Sentiment Analysis for Mental Health [Dataset].

[2] Tadesse, M. M., Lin, H., Xu, B., & Yang, L. (2019). Detection of Depression-Related Posts in Reddit Social Media Forum. IEEE Transactions on Computational Social Systems, 6(5), 957–968.

[3] Kumar, A., & Garg, A. (2022). Sentiment Analysis for Mental Health Monitoring Using Deep Learning. Journal of Ambient Intelligence and Humanized Computing, 13(4), 1899–1912.